

Customer Support Ticket Analysis

```
# Load the skimr package for data quality and exploratory data analysis
library(skimr)
```

```
# Read the cleaned data file
data <- read.csv("Customer_support_data_cleaned.csv")
```

```
# Generate a comprehensive overview of the dataset, including data types,
# missing values, summary statistics, and value distributions
skim(data)
```

Table 1: Data summary

Name	data
Number of rows	85907
Number of columns	25
Column type frequency:	
character	18
numeric	7
Group variables	
None	

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
unique_id	0	1	36	36	0	85907	0
channel_name	0	1	5	7	0	3	0
category	0	1	6	18	0	12	0
sub_category	0	1	5	32	0	57	0
order_id	0	1	0	36	18232	67676	0
order_date_time	0	1	0	14	68693	13767	0
issue_reported_at	0	1	12	13	0	30923	0
issue_responded	0	1	12	13	0	30262	0
survey_response_date	0	1	6	7	0	31	0
customer_city	0	1	0	19	68828	1783	0
product_category	0	1	0	27	68711	10	0
agent_name	0	1	7	28	0	1371	0
supervisor	0	1	8	16	0	40	0
manager	0	1	10	15	0	6	0
tenure_bucket	0	1	3	15	0	5	0
agent_shift	0	1	5	9	0	5	0
reported_month	0	1	4	6	0	2	0
customer_remarks	0	1	0	2672	57151	17659	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
---------------	-----------	---------------	-----	-----	-------	----------	------------

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
item_price	68701	0.2	5660.77	12825.73	0	392	979	2699.75	164999	
connected_handling_time	85665	0.0	462.40	246.30	0	293	427	592.25	1986	
csat_score	0	1.0	4.24	1.38	1	4	5	5.00	5	
reported_day	0	1.0	16.40	8.82	1	9	16	24.00	31	
reported_week	0	1.0	33.05	1.31	30	32	33	34.00	35	
reported_year	0	1.0	2023.00	0.00	2023	2023	2023	2023.00	2023	
response_time_minutes	0	1.0	136.89	592.53	-	2	5	35.00	5758	
					1437					

```
# Count the number of missing values in each column
colSums(is.na(data))
```

```
##           unique_id           channel_name           category
##              0              0              0
##      sub_category           order_id           order_date_time
##              0              0              0
##   issue_reported_at   issue_responded   survey_response_date
##              0              0              0
##      customer_city   product_category           item_price
##              0              0           68701
## connected_handling_time   agent_name           supervisor
##      85665              0              0
##      manager   tenure_bucket           agent_shift
##              0              0              0
##      csat_score   reported_day           reported_week
##              0              0              0
##   reported_month   reported_year   response_time_minutes
##              0              0              0
##   customer_remarks
##              0
```

```
# Count the number of duplicate rows in the dataset
sum(duplicated(data))
```

```
## [1] 0
```

```
# Count the number of negative response times
sum(data$response_time_minutes < 0, na.rm = TRUE)
```

```
## [1] 3128
```

```
# Display summary statistics for each variable, including minimum, maximum,
# quartiles, mean (numeric), and frequency counts (categorical)
summary(data)
```

```

## unique_id      channel_name      category      sub_category
## Length:85907   Length:85907      Length:85907   Length:85907
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
##
## order_id      order_date_time  issue_reported_at issue_responded
## Length:85907   Length:85907      Length:85907   Length:85907
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##
## survey_response_date customer_city      product_category      item_price
## Length:85907      Length:85907      Length:85907      Min.   :    0
## Class :character   Class :character   Class :character   1st Qu.:  392
## Mode  :character   Mode  :character   Mode  :character   Median :  979
##                                     Mean    : 5661
##                                     3rd Qu.: 2700
##                                     Max.    :164999
##                                     NA's    :68701
## connected_handling_time agent_name      supervisor
## Min.   :    0.0      Length:85907      Length:85907
## 1st Qu.: 293.0      Class :character   Class :character
## Median : 427.0      Mode  :character   Mode  :character
## Mean    : 462.4
## 3rd Qu.: 592.2
## Max.    :1986.0
## NA's    :85665
## manager      tenure_bucket      agent_shift      csat_score
## Length:85907   Length:85907      Length:85907      Min.   :1.000
## Class :character Class :character   Class :character   1st Qu.:4.000
## Mode  :character Mode  :character   Mode  :character   Median :5.000
##                                     Mean    :4.242
##                                     3rd Qu.:5.000
##                                     Max.    :5.000
##
## reported_day  reported_week  reported_month  reported_year
## Min.   : 1.0    Min.   :30.00    Length:85907    Min.   :2023
## 1st Qu.: 9.0    1st Qu.:32.00    Class :character 1st Qu.:2023
## Median :16.0    Median :33.00    Mode  :character Median :2023
## Mean    :16.4    Mean    :33.05                      Mean    :2023
## 3rd Qu.:24.0    3rd Qu.:34.00                      3rd Qu.:2023
## Max.    :31.0    Max.    :35.00                      Max.    :2023
##
## response_time_minutes customer_remarks
## Min.   :-1437.0      Length:85907
## 1st Qu.: 2.0         Class :character
## Median : 5.0         Mode  :character
## Mean    : 136.9
## 3rd Qu.: 35.0

```

Max. : 5758.0
##